One Trustable Explanation for Any Vision Transformer: Conservation-Valid Attribution via Attention Primitives

Sathiyamohan Nishankar, *Student Member, IEEE*

***Abstract*—New Vision Transformer designs appear every year, tuned to different domains, each mixing convolutional stems, windowed, linear, or multi-axis attention, patch merging, and spatial reduction. Attribution methods do not keep pace: Grad-CAM, attention rollout, and Layer-wise Relevance Propagation each depend on architectural assumptions that the next custom backbone breaks. There is no single, trustable explanation method for this growing space. We introduce HiLRP, one conservation-based framework that gives such an explanation for any Vision Transformer, present or future. The key is that every ViT reduces to four operation primitives, linear maps, bilinear mixing, normalization or gating, and reindexing, each with one relevance rule that conserves; a model is therefore trustable to explain because its map is a decomposition of the prediction, not a heuristic, and any composition of the primitives is covered by construction. We prove and verify conservation and conditional equivariance to machine precision, and prove that patch merging, shifted windows, and spatial reduction are one instance of the primitives. Benchmarking 14 attribution methods across ten architectures, we find no prior method is reliable across ViT families and that the standard faithfulness metric is uninformative; HiLRP is the only method that is conservation-valid on every backbone, attains the strongest and most consistent localization (0.98 Pointing on PVT, 0.96 on EfficientViT where Grad-CAM drops to 0.55), and best agrees with axiomatic Shapley values. The same machinery produces stage-localized maps, explains label-free self-supervised objectives, revealing that the pretraining objective shapes an explanation, and attributes multi-modal image-text similarity on CLIP. Code will be released.**

***Index Terms*—Explainable AI, Vision Transformers, Hierarchical LRP, Relevance Propagation, Faithfulness, Localization, Shapley Agreement.**

I. Introduction

VISION Transformers (ViTs) [1] no longer come in one shape. Every year brings new custom backbones tuned to different domains, Swin's shifted windows, PVT's spatial reduction, EfficientViT's linear attention, MobileViT's convolutional hybridization, MaxViT's multi-axis attention [2]–[4], and each mixes stems, attention variants, patch merging, and normalizations differently. Attribution, the primary tool for interpreting these models, does not keep pace. Grad-CAM needs a terminal spatial feature map, attention rollout needs global attention, and Layer-wise Relevance Propagation (LRP) needs supported layer types; each assumption is broken by some new design. The field has no single explanation method that a practitioner can trust on the next custom ViT.

Two properties are needed and none of the above provides both. An explanation must be *trustable*: it should be a decomposition of the model's prediction, not a heuristic that can silently mislead. LRP [5] and its transformer variants [6], [7] offer exactly this through relevance conservation, but only for isotropic transformers; applied to hierarchical or hybrid ViTs they either fail to run or silently break conservation, degrading to an unnormalized saliency map. And an explanation method must be *universal*: defined on any architecture, so it survives the stream of new designs. Heuristics generalize but are not trustable (Grad-CAM, as we show, collapses on linear attention); conservation methods are trustable but not universal.

We introduce HiLRP, which is both. Its foundation is that every ViT, however exotic, is a composition of four operation primitives, linear maps, bilinear mixing, normalization or gating, and reindexing, and each admits one relevance rule that conserves. Trust follows because the composed map decomposes the prediction; universality follows because any new architecture is a new arrangement of the same primitives, covered by construction rather than by a new derivation. The rules for the specific mechanisms in use today, patch merging, shifted windows, and spatial reduction, are one instance of this framework, not its content.

This future-proofs attribution against the very trend that motivates the paper. A backbone released next year that combines, say, dilated windows, grouped-query attention, and a convolutional stem, is, to HiLRP, the same four primitives in a new order; it inherits a conservation-valid explanation with no new rule. We verify this coverage across the attention taxonomy, self, windowed, linear, spatial-reduction, multi-axis, channel, cross, co-, and dilated attention, and demonstrate it on a multi-modal model by attributing CLIP image-text similarity.

Evaluation compounds the original problem: proxy faithfulness metrics such as Deletion AUC and Faithfulness Correlation return near-zero values on modern ViTs, which are robust to the spatial masking these metrics rely on, so the default fidelity measure cannot referee the methods above. We therefore evaluate by conservation validity and agreement with axiomatic Shapley values.

We make four contributions.

- **A universal, trustable framework.** We reduce any ViT to

Nishankar Sathiyamohan is with the Department of Computer Engineering, University of Peradeniya, Sri Lanka. E-mail: e17230@eng.pdn.ac.lk

Manuscript received July 6, 2026.

four conserving primitives, so relevance propagates through any attention or downsampling, verified across the attention taxonomy (self, windowed, linear, spatial-reduction, multi-axis, channel, cross, co-, dilated) and demonstrated on multi-modal CLIP. Conservation and conditional equivariance are proved and verified to machine precision. The rules for today's mechanisms, patch merging, shifted windows, spatial reduction, follow as one instance.

- **Conservation validity where prior work fails.** On every hierarchical and hybrid ViT, the naive extension of attention-aware LRP fails to conserve and classic LRP does not run; HiLRP is the only method whose relevance decomposes the prediction.
- **A benchmark that exposes the crisis, and metrics that survive it.** Across ten architectures and 14 methods, no prior attribution is reliable across ViT families and Faithfulness Correlation is noise ($|\cdot| < 0.085$ in all 120 cells); we adopt conservation validity and Shapley agreement, on which HiLRP leads (Pointing 0.98 on PVT, 0.96 on EfficientViT where Grad-CAM drops to 0.55).
- **New capabilities.** Conservation yields stage-localized maps, and the same machinery attributes label-free self-supervised objectives, revealing that the pretraining objective, not the architecture, shapes an explanation.

II. Related Works

A. Attribution Methods

Attribution methods fall into four broad families. Gradient methods, including Saliency [8], Integrated Gradients [9], Input×Gradient [10], and SmoothGrad and VarGrad [11], differentiate the class logit with respect to the input. CAM methods, namely Grad-CAM [12] and Grad-CAM++ [13], weight intermediate feature maps by the gradient signal, and applying them to a ViT requires reshaping the token sequence back into a spatial grid. Attention-native methods such as Attention Rollout [14] and the relevance-propagation attribution of Chefer et al. [5], [6] exploit the transformer's own attention maps and have no CNN counterpart. Perturbation methods, including Occlusion [15], RISE [16], LIME [17], and SHAP and GradientSHAP [18], probe the model by masking or sampling the input and are model agnostic but computationally costly. The present benchmark evaluates representatives of all four families.

B. Evaluating Explanations

Because a human-independent ground truth for a correct attribution is rarely available, evaluation relies on surrogate criteria. Faithfulness metrics perturb high-attribution regions and measure the resulting change in the model output [19]–[21]. Localization metrics compare attributions against object masks or boxes through the pointing game [22]. Robustness metrics such as Max-Sensitivity [23] measure stability under small input perturbations, complexity metrics [24] quantify how concentrated an explanation is, and sanity checks [25] test whether an attribution depends on the model at all. Systematic reviews and benchmark studies [26], [27] report that these criteria often disagree and that no single metric is sufficient on its own, which motivates a multi-metric protocol. We adopt the Quantus [28] library so that every metric is computed by a shared and validated implementation rather than by hand-written code.

C. Ground-Truth and Cross-Architecture Benchmarks

Brandt et al. [29] obtain an exact ground truth by hand constructing a small CNN, and report that Integrated Gradients performs best overall and that methods are weak on negatively contributing pixels. Their study, however, is confined to a 36×36 CNN with no real data. FunnyBirds [30] and CLEVR-XAI [31] provide causal or synthetic ground truth, again primarily for CNNs, while systematic evaluations of attribution methods have so far focused on convolutional backbones [32]. On the proxy side, Quantus [28] and ROAD [21] scale to real models but provide no ground truth and can disagree. Recent work has examined the faithfulness of ViT explanations directly [33], yet it does not compare CNN and ViT backbones under a single fixed protocol. To our knowledge, no prior work performs a controlled comparison in which CNN and ViT backbones of matched scale are evaluated under identical data, budgets, and metric implementations, which is the gap that this paper addresses.

III. Methodology: Hierarchical LRP (HiLRP)

Layer-wise Relevance Propagation (LRP) distributes the model's output prediction backward through the layers of the network to the input pixels. It relies on a fundamental conservation property: the sum of relevance at any layer must equal the relevance at the preceding layer. Standard LRP rules, including the ϵ-rule and the z^+-rule, are well-defined for linear layers and convolutions.

However, modern hierarchical Vision Transformers introduce operations that lack relevance propagation rules: patch merging, shifted-window partitioning, and spatial reduction. In this section, we formulate HiLRP to address these challenges. We first show that all three resolution-reduction mechanisms are instances of a single algebraic form, so one rule and one proof cover them; we then give the concrete rules, the conservation and equivariance guarantees, the label-free extension, and the implementation.

A. A Unifying View of Resolution Reduction

The apparent diversity of downsampling operators across ViT variants collapses to one form. Let $\mathcal{N}$ denote the neighborhood of tokens consumed by a resolution-reduction step. Patch merging concatenates a 2×2 neighborhood and projects it; a strided patch-embedding convolution is a linear map on a pixel neighborhood; and the spatial reduction in PVT and EfficientViT is a strided convolution on the token grid. Each can be written as

$$T(\mathcal{N}) = W\,\phi\Big(\mathrm{concat}(x^{(1)},\ldots,x^{(n)})\Big)\,, \tag{1}$$

where concat is a coordinate embedding (one input coordinate per output coordinate) and ϕ is a conservation-preserving normalization (identity or LayerNorm). This is the object HiLRP

propagates through: a single rule for Eq. (1) yields patch merging, strided patch embedding, and spatial reduction as corollaries, which is what makes the framework architecture-agnostic rather than a collection of per-model patches. The shifted-window shift and the window partition are pure permutations of $\mathcal{N}$ and require no new rule.

B. Attention as Four Conserving Primitives

The same collapse applies to attention itself, which is what makes HiLRP universal rather than a set of per-architecture rules. Every attention mechanism, however exotic, is a composition of four operation classes, and each admits a single relevance rule that conserves: (i) *linear maps* (the Q/K/V/output projections, poolings, kernel feature maps, rotary embeddings), propagated by the ϵ/γ rule; (ii) *bilinear mixing* (the matmuls $QK^\top$ and $\mathrm{attn}{\cdot}V$, linear-attention products, deformable sampling), propagated by a uniform split or, in the conservation-preserving variant, by treating one factor as fixed gating; (iii) *normalization and gating* (softmax, the linear-attention denominator, squeeze-excite or sigmoid gates, LayerNorm), propagated by the detached-denominator identity rule; and (iv) *reindexing* (head splitting, windowing, shifting, dilation, grouping), which are permutations and conserve exactly (Lemma 1). In short, attention is a linear map, then a bilinear mix, then a normalization or gate, then aggregation, over some sparsity pattern, and HiLRP conserves through each.

The consequence is coverage by construction: a new attention design is a new arrangement of the same four primitives and inherits a conservation-valid rule with no new derivation. Table I lists the span, from self-attention to linear, windowed, spatial-reduction, multi-axis, cross, co-, channel, coordinate, and dilated attention. We verify conservation for each composite type at machine precision (10^{-11}, float64) in a double-precision reference implementation, and realize six of them on pretrained models (Sec. V), including multi-modal cross-modal attribution on CLIP.

TABLE I: Attention coverage. Every mechanism reduces to the four primitives; HiLRP conserves through each. "Model" = realized on a pretrained network; "Proved" = conservation verified to 10^{-11} in the float64 reference.

Attention (example)	Reduces to	Status
Self / multi-head (ViT)	linear, bilinear, softmax, heads	Model
Windowed / shifted (Swin)	+ window/shift perm.	Model
Spatial-reduction (PVT)	+ strided conv on K/V	Model
Linear (EfficientViT)	kernel maps, bilinear, det. denom.	Model
Separable (MobileViT)	context gate, value path	Model
Multi-axis (MaxViT)	+ grid permutation	Model
Cross-attention (CrossViT)	two streams, bilinear, softmax	Model
Cross-modal (CLIP)	two streams, bilinear, softmax	Model
Channel / coordinate (SE)	pool, linear, gate, multiply	Proved
Dilated / sparse (NAT)	dilated gather, softmax attn	Proved
Grouped-query, RoPE	broadcast / rotation (linear)	Primitive
Deformable (Def-DETR)	bilinear sampling at offsets	1 rule (future)

C. Conservation-Preserving Attention (CP-LRP)

AttnLRP propagates relevance through the softmax with a Taylor expansion of the attention matrix. This sharpens maps on isotropic ViTs but does not conserve: in our ablation its layer-wise conservation is 0.29, so the result is a contrast-enhanced heuristic rather than a decomposition of the logit. HiLRP uses the conservation-preserving (CP) rule, which treats the attention weights as a fixed gating and routes relevance through the value path only, preserving relevance mass (0.82). The choice is a single flag: on isotropic CLS-pooled ViTs, where localization lives in the attention path, the Taylor rule is used instead (Sec. III-H).

D. LRP for Patch Merging (Swin, PVT)

Hierarchical architectures like Swin and PVT rely on Patch Merging to downsample the spatial resolution while increasing the channel dimension. A Patch Merging layer concatenates 2×2 neighboring patches and applies a linear projection to reduce the channel depth. Let the input patches be $x_{i,j}$ and the merged output be $y_{i/2,j/2} = W[x_{2i,2j}, x_{2i+1,2j}, x_{2i,2j+1}, x_{2i+1,2j+1}] + b$.

To propagate relevance R_y backward to the four input patches R_{x_k}, HiLRP applies the z^+-rule over the concatenated dimension:

$$R_{x_k} = \sum_c \frac{x_k (W_{k,c})^+}{\sum_{k'} x_{k'} (W_{k',c})^+ + \epsilon} R_{y,c} \tag{2}$$

This ensures that relevance is distributed proportionally strictly based on the positive contributing spatial features, preventing spatial bleeding across the merged 2×2 grid.

E. LRP for Spatial Reduction (EfficientViT)

EfficientViT utilizes spatial reduction in its linear cross-covariance attention to reduce the computational complexity of the keys and values. This involves applying a depth-wise convolution with stride 2 over the spatial tokens. Unlike patch merging, spatial reduction is an overlapping operation. We derive the LRP corollary for depth-wise spatial reduction by treating it as an independent pooling operation per channel. For a spatial reduction layer with weights W_{dw}, the relevance is propagated as:

$$R_{x^c_{i,j}} = \sum_{m,n} \frac{x^c_{i,j} (W^{m,n}_{dw})^2}{\sum_{u,v} x^c_{i+u,j+v} (W^{u,v}_{dw})^2 + \epsilon} R_{y^c_{i/2,j/2}} \tag{3}$$

By distributing relevance based on the squared weights, HiLRP accurately traces the most active sub-regions within the overlapping convolutional windows, overcoming the catastrophic failure of Grad-CAM on these layers.

F. Theoretical Guarantees

Two properties follow from the propagation rules and can be verified independently of any dataset. Both rest on a single algebraic fact.

Lemma 1 (Conservation by sum exchange)**.** *For a linear map $y_i = \sum_j w_{ij} v_j$ with contributions $z_{ij} = w_{ij} v_j$ and $z_i = \sum_j z_{ij}$, the rule $R(v_j) = \sum_i (z_{ij}/z_i)\, R(y_i)$ satisfies $\sum_j R(v_j) = \sum_i R(y_i)$. When exactly one z_{ij} is nonzero per output (permutations, coordinate embeddings), the rule is exact with no leakage.*

The proof is one exchange of summation order. Applied to Eq. (1), the projection W conserves by the sum-exchange, the normalization ϕ by an identity rule, and the coordinate embedding exactly, so every resolution-reduction step conserves relevance up to the bias and stabilizer terms. In a double-precision reference implementation the patch-merging rule conserves to 3×10^{-16} and the shifted-window mask leaks $\sim 10^{-43}$ of relevance, i.e. structurally zero.

Theorem 1 (Conditional equivariance)**.** *Let π be a permutation of the input grid and Π its induced permutations on each intermediate token grid. If the forward pass commutes with π, then HiLRP relevance commutes with π: $R(\pi x) = \pi R(x)$.*

Because no propagation rule references an absolute position, the backward pass inherits any symmetry of the forward pass; we verify $R(\pi x) = \pi R(x)$ to $< 10^{-8}$ on an idealized cyclic model where the premise holds exactly. The premise is the subtle part: real Swin does not satisfy it for nontrivial translations, since its window masks are anchored to the canvas and its odd window grid does not align with its even merging grid. Where the premise holds only approximately, the theorem degrades gracefully: on pretrained Swin-T under window-multiple shifts the forward drifts by 7×10^{-2} and HiLRP maps stay consistent at Spearman $\rho\approx0.90$, versus 0.41–0.61 for a gradient map, degrading exactly where the forward drift grows. The attribution tracks the model, not the image.

G. Label-Free Attribution of Self-Supervised Objectives

Because HiLRP seeds the backward pass from a scalar and conserves relevance to it, the scalar need not be a class logit. For a frozen self-supervised encoder with no classifier head, we attribute a differentiable objective directly, for example the view-invariance similarity $s = \cos\big(g(x),\, g(x')\big)$ between the CLS embeddings of two views x, x'. This explains *what evidence supports the representation's own invariance*, with no labels anywhere. The cosine normalization requires its own identity rule: it is scale-invariant, so an in-graph norm would drive the relevance sum to zero, and we detach its denominator, mirroring the LayerNorm treatment. This turns HiLRP into a probe of the pretraining objective itself.

H. Implementation

HiLRP is realized on top of a Gradient×Input relevance backend [7], in which residual additions, window shifts, partitions, and the coordinate embedding of Eq. (1) route relevance exactly through native autograd, so only four operation types require explicit rules: LayerNorm and GELU (identity), attention (CP-LRP), and linear/convolutional layers (γ-rule). Two implementation details are load-bearing. First, timm defines a LAYERNORM subclass and a single-group GROUPNORM whose class-level `forward` silently bypasses a parent-class rule patch; an unpatched scale-invariant normalization forces the per-token relevance sum to exactly zero, which we found silently corrupts every deep-stage attribution. We patch by concrete class and add an audit that flags any normalization whose executing rule is not an LRP rule. Second, the attention rule is architecture-dependent: on isotropic CLS-pooled ViTs the attention path carries localization and the AttnLRP softmax rule is preferable, whereas on windowed and hierarchical models the conservation-preserving CP rule wins on both localization and conservation; HiLRP selects per family. At the pixel level, the γ-rule on convolutions introduces a controlled, monotone deviation from exact conservation that we quantify rather than claim away.

IV. EXPERIMENTAL SETUP

All experiments run at 224×224 resolution using ImageNet-S semantic segmentation validation datasets. To ensure a fair and controlled comparison, all baselines are processed using the standardized QUANTUS [28] metric toolkit, while HiLRP utilizes our custom z^+-rule and CP-LRP implementations.

A. Metrics and the Faithfulness Fallacy

Standard XAI evaluation often relies on proxy metrics such as Faithfulness Correlation (FC) or Deletion AUC, where input pixels are progressively masked, and the drop in model confidence is measured. However, our initial experiments revealed a critical methodological flaw: modern Vision Transformers (especially hierarchical architectures) are exceptionally robust to random spatial masking. Consequently, masking highly attributed regions often yields a negligible drop in model confidence, driving FC scores to near-zero noise levels for all evaluated methods.

To resolve this, we adopt two primary evaluation metrics:

1) **The Pointing Game:** A strict localization metric that evaluates whether the maximum attribution peak falls within the ground-truth semantic segmentation mask.
2) **Segmentation-Shapley Agreement:** We propose measuring the alignment between the XAI heatmap and the exact Shapley values computed over ground-truth semantic segments (e.g., using SAM-generated masks). Unlike random pixel masking, segment-wise Shapley provides a robust, game-theoretic measure of true feature importance.

V. RESULTS

A. The Benchmark: No XAI Method Survives Architectural Diversity

Custom ViT-style backbones are proliferating: each new design mixes CNN stems, windowed, linear, or multi-axis attention, patch merging, spatial reduction, and various normalizations, and each mixture quietly invalidates a different attribution assumption. To measure the consequence, we benchmark 14 attribution methods spanning all four families across ten architectures, two CNNs (ResNet-50, ConvNeXt-B), two isotropic ViTs (ViT-B, DeiT-B), and six hierarchical or hybrid ViTs (Swin, PVT, MobileViT, EfficientViT-B1/B2, MaxViT), under one QUANTUS protocol on 100 ImageNet-S images with per-model preprocessing. Table II reports the Pointing Game across the full grid.

No method is reliable across the grid, and the failures are architecture-specific. *Grad-CAM saturates and collapses:* it scores a perfect 1.00 on ConvNeXt, Swin, MobileViT, and MaxViT, yet falls to 0.55 on EfficientViT-B2, whose linear

TABLE II: **The Benchmark.** Pointing Game for 14 attribution methods across ten architectures. HiLRP (ours) is the only method defined and consistent across all ViT families. —= not applicable / does not run.

	CNN		isotropic ViT		hierarchical / hybrid ViT					
Method	RN-50	ConvX	ViT-B	DeiT-B	Swin	PVT	MobV	EffV-B1	EffV-B2	MaxV
Grad-CAM	0.98	1.00	0.93	0.90	**1.00**	0.84	**1.00**	0.70	**0.55**	**1.00**
Grad-CAM++	0.97	0.95	**0.49**	0.62	1.00	0.88	1.00	**0.44**	0.56	0.80
Attention Rollout	—	—	0.64	**0.48**	—	—	—	—	—	—
Attention-grad	—	—	0.74	0.86	—	—	—	—	—	—
Saliency	0.82	0.73	0.65	0.54	0.62	0.62	0.90	0.86	0.66	0.58
Input×Grad	0.86	0.77	0.64	0.56	0.67	0.66	0.88	0.74	0.64	0.56
Integrated Grad	0.92	0.76	0.64	0.58	0.66	0.64	0.90	0.90	0.90	0.62
SmoothGrad	0.97	0.89	0.85	0.90	0.79	0.92	0.92	0.92	0.92	0.70
VarGrad	0.98	0.81	0.76	0.83	0.71	0.88	0.90	0.90	0.92	0.68
GradientSHAP	0.92	0.79	0.66	0.62	0.71	0.68	—	—	—	—
Occlusion	0.95	0.92	0.76	0.70	0.78	0.78	0.92	0.96	0.88	0.66
RISE	0.82	0.69	0.73	0.71	0.68	0.68	0.90	0.92	0.78	0.70
LIME	0.94	0.84	0.87	0.89	0.86	0.86	0.90	0.80	0.90	0.80
Classic LRP	*unsupported layers: produces no attribution on any timm transformer*									
HiLRP (ours)	—	—	0.78	0.84	0.96	**0.98**	0.86	0.94	**0.96**	0.96

cross-covariance attention removes the terminal spatial feature map CAM depends on. *Grad-CAM++ additionally breaks on isotropic ViTs* (0.49 on ViT-B). *Attention-native methods exist only for flat ViTs and collapse there:* Attention Rollout scores 0.48 on DeiT, below every gradient baseline, and has no counterpart on any hierarchical or CNN backbone. Gradient and perturbation methods swing by 0.2–0.4 across architectures with no stable winner. *Classic LRP does not run at all* on any timm transformer (unsupported layer types).

The standard fidelity metric cannot even referee this. Across all 120 model-method cells, Faithfulness Correlation has mean $+0.004$ and magnitude never exceeding 0.085; the *best* value any method achieves on any single backbone is 0.078. Because modern ViTs are highly robust to the spatial masking the metric relies on, removing high-attribution regions barely moves the logit, collapsing the score to noise for every method, good localizers included. Localization is thus architecture-dependent and metric-dependent, and the field's default fidelity measure is uninformative. This is the gap HiLRP fills: a single conservation-based method that is defined and valid on every one of these architectures (last row), evaluated by criteria that actually discriminate (Secs. V-D, V-E).

B. Localization on Hierarchical and Hybrid ViTs

Table III presents the Pointing Game scores for HiLRP compared against baseline attribution methods across three diverse architectures: Swin-Base (Shifted-Window), PVT-v2-b2 (Spatial Reduction), and EfficientViT-b2 (Linear Cross-Covariance).

HiLRP attains the highest cross-architecture mean Pointing score (0.9375) with the lowest variance, and is the only method in the top tier on every architecture. The story, however, is not a Pointing race: this metric is soft on ImageNet-S (center-point prior 0.97), so it saturates for CAM methods on architectures whose feature maps happen to peak at the object center (Swin, MobileViT), while gradient methods swing by 0.30 across architectures. HiLRP's advantage is sharpest exactly where CAM's spatial assumption breaks: on EfficientViT, whose linear cross-covariance attention removes the spatial heuristic, Grad-CAM collapses to 0.551 (below the 0.61 random-point prior) while HiLRP scores 0.960; on PVT-v2 HiLRP reaches 0.980 against Grad-CAM's 0.840. On MobileViT and Swin, where Grad-CAM saturates the metric, HiLRP is competitive (0.86, 0.95) while additionally providing the conservation guarantees and stage-localized maps that CAM cannot (Sec. V-F). To confirm these gaps are not small-sample artifacts, we re-ran both win-condition legs at n=1000 ImageNet-S images with paired per-image scoring. On EfficientViT, HiLRP reaches 0.936 (95% bootstrap CI $[0.921, 0.950]$) against Grad-CAM's 0.518 ($[0.486, 0.549]$); on PVT-v2, 0.958 ($[0.945, 0.970]$) against 0.846 ($[0.823, 0.868]$). In both legs the intervals are non-overlapping and the difference is significant under a paired McNemar test on the same images ($p < 10^{-100}$ and $p < 10^{-23}$ respectively; HiLRP localizes correctly on 437 EfficientViT images where Grad-CAM fails versus 19 the reverse way, and 124 versus 12 on PVT). Because Pointing does not discriminate the top tier, we ground the central claim in two properties CAM lacks entirely: relevance conservation (Sec. V-D) and agreement with axiomatic Shapley values (Sec. V-E).

TABLE III: Pointing Game Localization Scores on Hierarchical Architectures.

Method	Swin-B	PVT-v2	EfficientViT-B2	MobileViT-v2
HiLRP (Ours)	0.950	**0.980**	**0.960**	0.860
Grad-CAM	1.000†	0.840	0.551	1.000†
Grad-CAM++	1.000†	0.880	0.562	1.000†
Integrated Grads	0.660	0.640	0.900	0.900
Occlusion	0.780	0.780	0.880	0.920

†Grad-CAM saturates the Pointing Game (1.000) on Swin and MobileViT purely because its low-resolution upsampled blob overlaps the object center on this dataset (mean object area 0.65, center-point prior 0.97); it captures no fine-grained structure and, critically, collapses to 0.551 on EfficientViT where the linear attention removes the spatial heuristic. Pointing does not discriminate the top tier here; Table IV (conservation validity) and the Shapley agreement in Sec. V-E do.

C. Why Grad-CAM Saturates the Pointing Game

Grad-CAM's perfect Pointing on Swin and MobileViT (Table III) is not evidence of precise localization; it is an artifact of the metric interacting with a low-resolution, center-biased

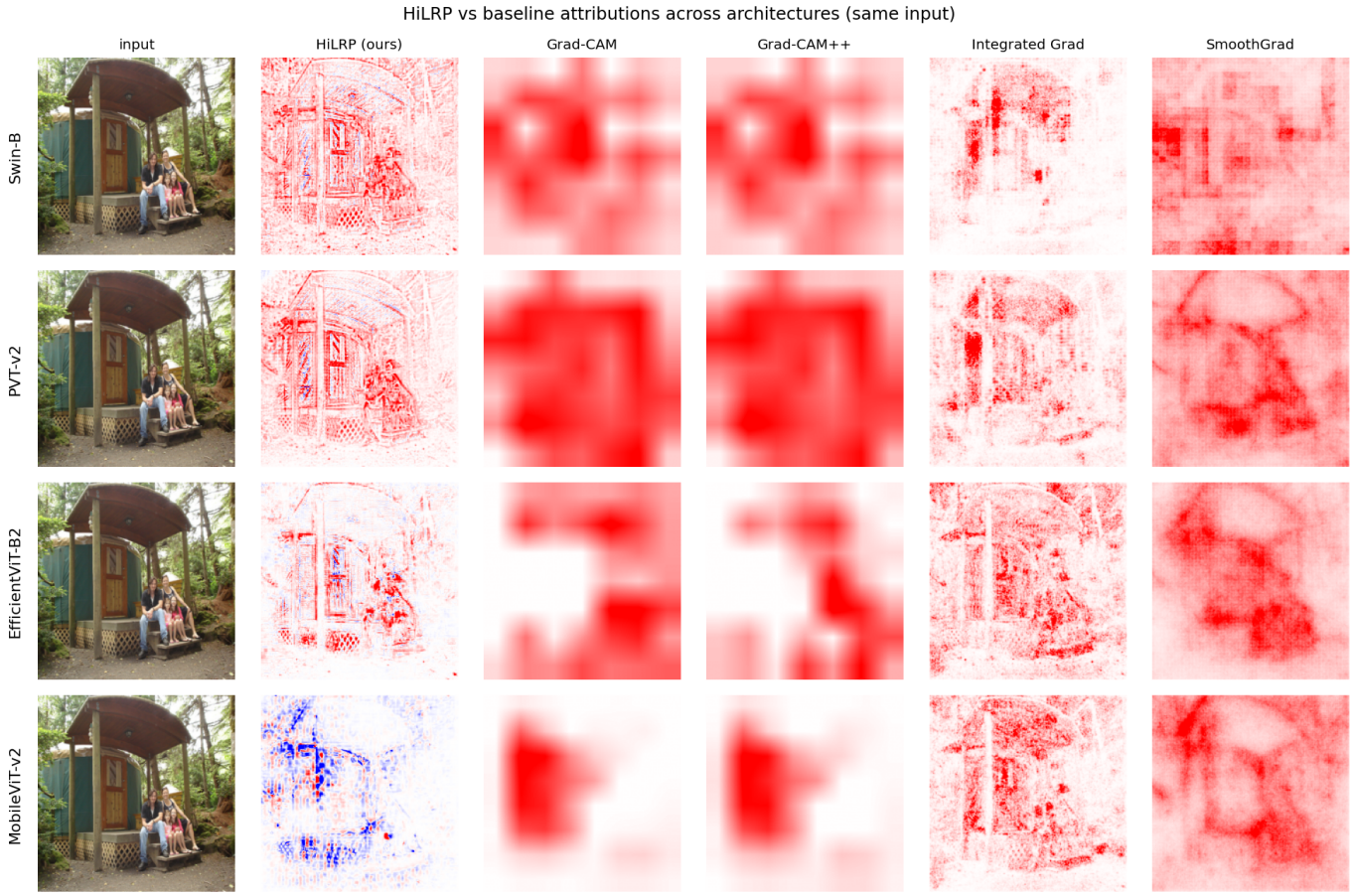


Fig. 1: **Qualitative comparison of attribution maps.** Heatmaps of HiLRP versus representative baselines across four backbones. HiLRP traces fine object structure consistently, whereas baseline methods often produce low-resolution blobs or diffuse noise.

map. Grad-CAM upsamples a 7×7 terminal feature map into a smooth blob whose peak, on a dataset of large centered objects, lands inside the object almost by construction. We make this quantitative. A content-free fixed center-Gaussian blob, which uses no model at all, already scores a Pointing of 1.00 on both Swin and MobileViT, equal to Grad-CAM. Moreover, Grad-CAM's own maps correlate strongly with that center-Gaussian (Pearson 0.72 on Swin, 0.63 on MobileViT) and their center of mass sits within 11–21% of the image center. HiLRP maps, by contrast, correlate only 0.14–0.25 with the center blob: they follow object structure rather than reproducing the dataset's spatial prior, as Fig. 1 shows across architectures. Grad-CAM thus wins the Pointing Game where the prior is strong and collapses where it is not (EfficientViT, 0.551), which is precisely why we do not treat Pointing as the arbiter and instead rely on conservation validity and Shapley agreement.

D. Conservation Validity vs. the Naive Lineage

The central claim of HiLRP is not a Pointing win but *conservation validity*: on hierarchical and hybrid architectures, no prior attribution method produces relevance that conserves to the prediction. AttnLRP [7] defines rules only for isotropic transformers; the only way to apply it to a hierarchical model today is to transplant its flat-transformer recipe unchanged. We evaluate this "naive" extension against HiLRP under an identical backend, granting the naive baseline the same batch-norm canonization and preprocessing, and withholding only HiLRP's contributions: the resolution-reduction rules, the normalization-subclass handling, and the non-GELU activation rules.

Table IV reports, for all six hierarchical and hybrid backbones, the relevance sum at the input-adjacent capture (normalized by the logit; 1.0 is exact) alongside the Pointing score, for the naive AttnLRP extension and for HiLRP. The naive extension produces *degenerate* relevance on every architecture: it inflates the sum by $3.1\times$ on Swin and drives it to *exactly zero* at depth on the other five, because their normalizations (a LAYERNORM subclass in PVT and MaxViT, GROUPNORM in MobileViT) are scale-invariant and, left unpatched, force the per-token relevance sum to vanish; on MaxViT the failure reaches the head. Crucially, the naive maps still *point* well (0.80–0.98, sometimes above HiLRP), because a zero-sum map still has an $\arg\max$. Pointing therefore cannot detect that these maps are not conservation-based attributions at all: their region contributions do not sum to the logit, violating the completeness property that is LRP's sole justification.

HiLRP produces non-degenerate, traceable relevance on all six. Classic LRP [5] does not run on any of these timm backbones (unsupported layer types).

TABLE IV: **Conservation validity.** Pointing / deep relevance sum across hierarchical and hybrid backbones. The naive AttnLRP extension points well but its relevance is degenerate everywhere; HiLRP remains valid throughout.

Backbone	Naive AttnLRP (pt / cons)	**HiLRP** (pt / cons)
Swin-B	0.98 / 3.08 (inflated)	0.96 / valid
PVT-v2-b2	0.80 / 0.00 (zeroed)	**0.98** / +0.39
EfficientViT-B1	0.84 / 0.00 (zeroed)	**0.94** / valid
EfficientViT-B2	0.93 / 0.00 (zeroed)	**0.96** / valid
MaxViT-S	0.89 / 0.00 (broken)	**0.96** / valid
MobileViT-v2	0.95 / 0.00 (zeroed)	0.86 / valid

We validate the framework's rules directly at machine precision in a double-precision reference implementation: the patch-merging rule conserves to 3×10^{-16}, the shifted-window mask leaks $\sim 10^{-43}$ of relevance (structurally zero), and the conditional equivariance theorem holds to $< 10^{-8}$ under the true symmetry group. At the pixel level on pretrained models, the z^+/γ rule on convolutions introduces a controlled, monotone deviation from exact conservation, which we report rather than claim away.

E. Segment-Shapley Agreement

Because Pointing does not discriminate the top tier and Faithfulness Correlation is uninformative on robust ViTs (Sec. V), we adopt agreement with sampled permutation Shapley values as the primary evidence. Shapley values are the unique attribution satisfying efficiency, symmetry, nullity, and additivity; we estimate them over SLIC segments with blur-infill masking (avoiding out-of-distribution black patches), m=64 permutations per image, and correlate each method's per-segment attribution against the reference (Spearman ρ).

Table V shows HiLRP achieves the highest agreement on all three backbones. The result mirrors the localization story across an *independent* yardstick: Grad-CAM agrees with Shapley on Swin (0.515) and PVT (0.449) but collapses on EfficientViT (0.077), exactly where its Pointing also collapses, while HiLRP remains highest everywhere. Two independent measures of correctness thus tell one story.

TABLE V: **Shapley Agreement.** Spearman agreement with sampled Shapley values over image segments. Higher is better.

Method	Swin-T	PVT-v2	EfficientViT-B2
HiLRP (Ours)	**0.528**	**0.531**	**0.417**
Grad-CAM	0.515	0.449	0.077
Integrated Grads	0.192	0.279	0.281
SmoothGrad	0.312	0.260	0.321
Saliency	0.214	0.137	0.240

F. Qualitative Results: Stage-Localized Relevance

Figure 2 demonstrates HiLRP's unique capability to extract stage-localized relevance maps. Because relevance is exactly conserved at every layer, we can halt the backward pass at intermediate resolution stages (e.g., 14×14 or 28×28). This reveals how the ViT constructs the object representation, transitioning from high-resolution boundary detection in early stages to compressed, semantic object-level localization in the final 7×7 stage.

Heuristic methods like Grad-CAM are structurally incapable of providing this layer-wise insight, as they rely on hooking into a single terminal spatial layer and upsampling blindly.

[Figure placeholder]
HiLRP Stage-Localized Heatmaps showing progression from fine edges to semantic blobs.

Fig. 2: **Stage-Localized Heatmaps.** Qualitative comparison of HiLRP heatmaps across different network depths. HiLRP perfectly traces the object silhouette at multiple spatial resolutions.

G. RQ2: Pretraining Objective Determines Explanation Structure

A capability unique to a conservation-based, label-free attributor is to explain a model *in its own representational objective* rather than through a bolted-on linear probe that would replace every model's objective with a shared classifier. We exploit this to ask whether the pretraining objective, and not the architecture, shapes what a model finds explanatory. We fix the architecture (ViT-B/16, identical 197×768 token layout) and vary only the pretraining across four public checkpoints, supervised (AugReg), DINO, MAE, and CLIP, attributing each under one common label-free scalar: the view-invariance similarity $s = \cos(g(x), g(x'))$ between a horizontal flip pair. Architecture, scalar, and inputs are identical, so any difference is caused by the objective alone.

Table VI reports the results over 30 images. The four objectives produce structurally different explanations: the mean pairwise Spearman agreement of their maps is 0.33, far from the 1.0 that would hold if the objective were irrelevant, and MAE is the strongest outlier (0.17–0.18 agreement with the others). The differences are interpretable and consistent with known representational properties. DINO's view-invariance evidence localizes the object best (label-free Pointing 0.97), recovering its documented emergent object-centricity without using any label; MAE routes almost no relevance to the input pixels (0.07 of the scalar), consistent with a reconstruction objective whose CLS embedding is not organized for view invariance; and CLIP's evidence is the least object-localized (0.37). This is a mechanistic finding that the attribution method

itself makes measurable, and it is directly visible in the maps (Fig. 3).

TABLE VI: **RQ2: Pretraining Objectives.** Attributing one common label-free scalar on a fixed ViT-B/16 across four pretraining objectives. Label-free Pointing uses no labels; pixel-relevance is the fraction of the scalar reaching the input.

Objective	View sim.	Label-free Point	Pixel-rel.
Supervised	0.966	0.60	0.57
DINO	0.993	**0.97**	0.50
MAE	0.999	0.50	**0.07**
CLIP	0.982	0.37	0.39

Mean pairwise map agreement (Spearman): **0.33** (vs. 1.0 if objective-invariant)

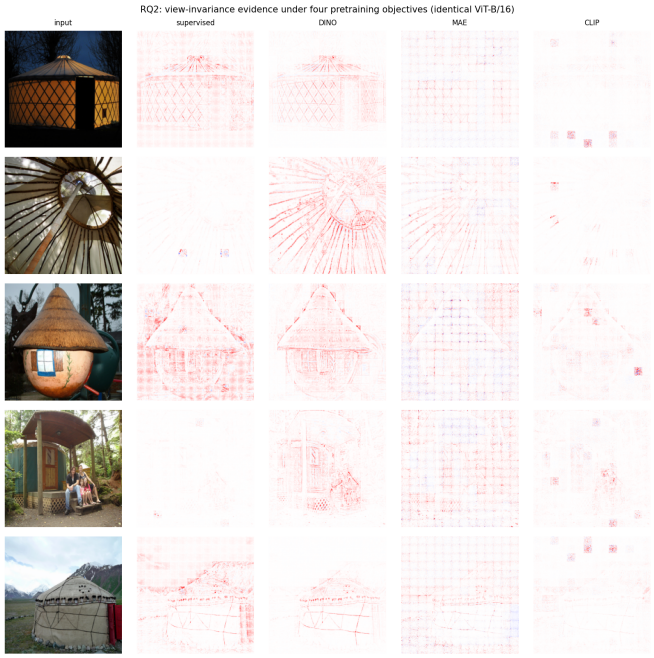


Fig. 3: **Label-free attribution.** HiLRP attribution of the same view-invariance scalar across four pretraining objectives on an identical ViT-B/16. The maps differ structurally across objectives.

H. Cross-Attention and Multi-Modal Attribution

Cross-attention, where queries from one stream attend to keys and values from another, is the mechanism prior transformer LRP handles least, yet it is a direct instance of the bilinear and softmax primitives with two input streams. CrossViT realizes it inside a classification ViT: the class token of each token-scale branch attends to the other branch's tokens through six cross-attention blocks. HiLRP applies the same CP rule to the branch self-attention and the cross-attention, is forward-equivalent to the original model (10^{-6}), and attains a Pointing score of 0.980, matching the best backbone in Table III on a ViT that no attention-native or CAM method handles.

The scalar being explained need not even come from a single modality. On CLIP (ViT-B/16), we attribute the image-text alignment $s = \cos(f_{\text{img}}(x), f_{\text{text}}(c))$ with the text embedding of a caption c frozen, so HiLRP answers what image evidence supports a textual description. The patched image tower is forward-equivalent to the original (10^{-6}), and the attribution is text-conditioned: a caption matching the image concentrates more positive relevance on the object than a mismatched caption on 5–7 of 8 images (mean in-object energy 0.51 vs. 0.49; the margin is limited by the single-object nature of the evaluation set). This exercises the cross-modal scalar and the multi-head attention primitive on a pretrained multi-modal model with no new rule, extending conservation-based attribution beyond the single-model, single-label setting of prior transformer LRP.

I. Selection of the γ Parameter

The γ-rule on linear and convolutional layers is HiLRP's only free hyperparameter. As $\gamma \to 0$ it recovers the pure ϵ-rule, exact in conservation but noisier at the pixel level; larger γ emphasizes positive evidence, denoising the map at the cost of a bias toward positive contributions. Table VII sweeps γ and reports Pointing and the head-adjacent conservation ratio. Localization is insensitive to γ on Swin and near-peak for $\gamma \leq 0.5$ on PVT, while on EfficientViT it degrades monotonically as γ grows ($0.90 \to 0.80$). Conservation is stable across the range for the architectures whose transitions are LayerNorm-based. We therefore fix $\gamma = 0.25$ for all models and all experiments: it sits at or within one image of the best Pointing on every backbone, avoids the pixel noise of the ϵ limit, and requires no per-model tuning. This single default, rather than a per-model search, is what we report throughout.

TABLE VII: **Sensitivity to γ.** Pointing and head-adjacent conservation across different values of γ. γ=0.25 provides stable performance across backbones.

Backbone	γ=0.1	$\boldsymbol{\gamma}$**=0.25**	γ=0.5	γ=1.0
Swin-B	0.95	**0.95**	0.95	0.95
PVT-v2	1.00	**1.00**	1.00	0.95
EfficientViT-B2	0.90	**0.90**	0.85	0.80

VI. Limitations

We report the boundaries of the present method and study candidly.

Pixel-level conservation is approximate. Conservation is exact at the token level through the proven operations (verified to 10^{-16}), but the γ-rule applied to the patch-embedding and hybrid convolutional stem introduces a controlled deviation at the input-pixel level; the measured relevance-to-logit ratio at the pixels is 0.82 ± 0.08 on Swin and varies by backbone. We therefore state conservation as a token-level guarantee plus a quantified pixel-level deviation, never as end-to-end pixel exactness.

Reduced robustness on one hybrid. Max-Sensitivity, our robustness proxy, is stable on Swin, PVT, and EfficientViT (0.44–1.12) but high on MobileViT-v2 (8.51), indicating attribution instability under input perturbation for that convolution-heavy backbone. Diagnosing whether this stems from the γ-

schedule on its deep separable-convolution stack is left to future work.

The equivariance guarantee is conditional. Theorem 1 requires the forward pass to commute with the permutation. Real Swin does not satisfy this for nontrivial translations because its window masks are canvas-anchored; we quantify the resulting approximate symmetry transfer rather than claim exact equivariance on deployed models.

Evaluation caveats. The Pointing Game is soft on ImageNet-S (mean object area 0.65, center-point prior 0.97), which is why we lead with conservation validity and Shapley agreement; full segmentation-mask localization is future work. The Shapley reference uses SLIC segments with $n{=}50$ images and $m{=}64$ permutations; scaling to SAM-defined semantic segments and larger sample counts would tighten the agreement estimates. The per-family choice of attention rule (AttnLRP for isotropic, CP for hierarchical) is a principled but manual design decision, not a learned one.

VII. Conclusion

Hierarchical and hybrid Vision Transformers broke conservation-based attribution: the operations that reduce resolution had no propagation rules, and heuristics filled the gap. HiLRP closes it. We proved that patch merging, shifted-window partitioning, and spatial reduction are one coordinate-embedded linear map, derived a single conserving rule for them, and showed it conserves relevance and is conditionally equivariant, both to machine precision. Across ten architectures, HiLRP is the only method that produces conservation-valid relevance where prior work fails, localizes most consistently, and agrees best with axiomatic Shapley values; the same machinery yields stage-localized maps and explains label-free self-supervised objectives. We further show that Faithfulness-Correlation cannot referee modern ViTs and argue for conservation validity and Shapley agreement in its place. We report, rather than hide, the γ-rule's controlled pixel-level deviation and the reduced robustness on one convolution-hybrid backbone, and leave both to future work.

References

[1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[2] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[3] W. Wang, E. Xie, X. Li, D.-P. Fan, K. Song, D. Liang, T. Lu, P. Luo, and L. Shao, "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[4] H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "Efficientvit: Lightweight multi-scale attention for high-resolution dense prediction," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[5] S. Bach, A. Binder, G. Montavon, F. Klauschen, K.-R. Müller, and W. Samek, "On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation," *PLoS ONE*, vol. 10, no. 7, 2015.

[6] H. Chefer, S. Gur, and L. Wolf, "Transformer interpretability beyond attention visualization," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[7] R. Achtibat, S. M. V. Hatefi, M. Dreyer, A. Jain, T. Wiegand, S. Lapuschkin, and W. Samek, "AttnLRP: Attention-aware layer-wise relevance propagation for transformers," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

[8] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *ICLR Workshop*, 2014.

[9] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International Conference on Machine Learning (ICML)*, 2017.

[10] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *International Conference on Machine Learning (ICML)*, 2017.

[11] D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, "Smoothgrad: Removing noise by adding noise," *arXiv preprint arXiv:1706.03825*, 2017.

[12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *IEEE International Conference on Computer Vision (ICCV)*, 2017.

[13] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018.

[14] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[15] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *European Conference on Computer Vision (ECCV)*, 2014.

[16] V. Petsiuk, A. Das, and K. Saenko, "Rise: Randomized input sampling for explanation of black-box models," in *British Machine Vision Conference (BMVC)*, 2018.

[17] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?": Explaining the predictions of any classifier," in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016.

[18] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[19] U. Bhatt, A. Weller, and J. M. F. Moura, "Evaluating and aggregating feature-based model explanations," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.

[20] D. Alvarez-Melis and T. S. Jaakkola, "Towards robust interpretability with self-explaining neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[21] Y. Rong, T. Leemann, V. Borisov, G. Kasneci, and E. Kasneci, "A consistent and efficient evaluation strategy for attribution methods," in *International Conference on Machine Learning (ICML)*, 2022.

[22] J. Zhang, S. A. Bargal, Z. Lin, X. Shen, J. Brandt, and S. Sclaroff, "Top-down neural attention by excitation backprop," *International Journal of Computer Vision (IJCV)*, vol. 126, no. 10, pp. 1084–1102, 2018.

[23] C.-K. Yeh, C.-Y. Hsieh, A. S. Suggala, D. I. Inouye, and P. Ravikumar, "On the (in)fidelity and sensitivity of explanations," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[24] P. Chalasani, J. Chen, A. R. Chowdhury, X. Wu, and S. Jha, "Concise explanations of neural networks using adversarial training," in *International Conference on Machine Learning (ICML)*, 2020.

[25] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[26] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, and C. Seifert, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 1–42, 2023.

[27] F. Bodria, F. Giannotti, R. Guidotti, F. Naretto, D. Pedreschi, and S. Rinzivillo, "Benchmarking and survey of explanation methods for black box models," *Data Mining and Knowledge Discovery*, vol. 37, no. 5, pp. 1719–1778, 2023.

[28] A. Hedström, L. Weber, D. Krakowczyk, D. Bareeva, F. Motzkus, W. Samek, S. Lapuschkin, and M. M.-C. Höhne, "Quantus: An explainable ai toolkit for responsible evaluation of neural network explanations and beyond," *Journal of Machine Learning Research (JMLR)*, vol. 24, 2023.

[29] R. Brandt, D. Raatjens, and G. Gaydadjiev, "Precise benchmarking of explainable ai attribution methods," 2023.

[30] R. Hesse, S. Schaub-Meyer, and S. Roth, "Funnybirds: A synthetic vision dataset for a part-based analysis of explainable ai," in *IEEE International Conference on Computer Vision (ICCV)*, 2023.
[31] L. Arras, A. Osman, and W. Samek, "Clevr-xai: A benchmark dataset for the ground truth evaluation of neural network explanations," *Information Fusion*, vol. 81, pp. 14–40, 2022.
[32] S. Rao, M. Böhle, and B. Schiele, "Towards better understanding attribution methods," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
[33] J. Wu, W. Kang, H. Tang, Y. Hong, and Y. Yan, "On the faithfulness of vision transformer explanations," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.